Winston Li

Updated on Oct 13, 2026.

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Education

UC Irvine • B.S. in Computer Science w/ Honors, 3.9 / 4.0 GPA — Sep 2024 – Jun 2026

- graduated in 2 years • ICS Honors Program • UROP Fellow
- Data Structures • Algorithms • Software Eng. • Operating Systems • Compilers
- Cryptography • Formal Lang. & Automata • Quantum Computation • Distributed Systems

Projects

manifold • *Rust, wgpu, CEF, Servo* — Sep 2026 – Present

- Built an 85k-line browser: an engine process per tab (Blink, Servo, WebKit) over IPC, with hot-swap.
- Added crash recovery, journaled sessions, and a force-directed tab graph on **wgpu**.

scratchtape • *Rust, wgpu, cubecl* — Sep 2026 – Present

- Built an ML engine from scratch (autodiff, layers, GPU backend); every primitive is gradient-checked.
- Measured **DX12** at 1.6–5.8× lower overhead than Vulkan on small matmuls; up to 900 GFLOP/s.

Languages & compilers • *Rust, C99, WebAssembly* — Feb 2023 – Present

- **tatic**: **JIT** caching **WebAssembly** by content hash; installs code only if it matches an interpreter and has a **machine-checked proof** of the compiler model; the rest stays interpreted.
- Fuzzing and benchmarks caught a checker overflow and a memory leak; no mismatches in ~12k trials.
- **cereal**: C99 front end and **language server**; matched gcc-13 on 3,908/3,910 runnable gcc.dg tests.
- **Lambast** (Honors): optimizing untyped λ-calculus compiler with **closure conversion**; UROP talk.
- **CS 142B project**: SSA compiler via Braun et al.'s **marker algorithm**; CSE, phi simplification.
- **renno**: CEK-machine language with **algebraic effects** and gradual typing; 400+ tests.
- **renno (prototype)** (2023): Rust compiler to JavaScript with name resolution and a type checker.

plumb • *Rust, Windows, NTFS* — Oct 2026 – Present

- Built a disk-usage analyzer that reads the **NTFS MFT** to scan a 4.3M-node volume in under 10 s.
- Added live updates from the NTFS change journal and a partial tree while scanning; 200+ tests.

Apps & web • *TypeScript, Svelte, Tauri* — Sep 2026 – Present

- **arrow-game**: strict-TS roguelite; 400+ tests; balanced with an evolved bot that lasts 3× longer.

Skills and interests

Proficient in Rust, C / C++, JS / TS, Python, Svelte, Tailwind.

Familiar with Git, HTML / CSS, Nix, LaTeX, Typst, Claude Code, Unix systems & shells.

Acquainted with most mainstream languages, e.g., C#, Go, Haskell, Java, Lua, Ruby, SQL.

Experience

Learning Assistant • UC Irvine — Mar 2025 – Jun 2026

- Delivered supplementary pedagogy for **Python** and upper-div **Algorithms,** e.g., Master Theorem.
- Spearheaded study sessions to identify and guide students through gaps in understanding.

Intern • LGBTQIA+ Identity Commission @ UC Irvine — Oct 2024 – Jun 2025

- Led a team of 3 to organize and host themed Gayme Show™.

Programmer • Video Game Development Club @ UC Irvine — Sep 2024 – May 2025

- Worked with team of 13 to create typing game in **C#** and **Unity**; led dev of the main UI window.